

## 4.4 Orthogonal Polynomials

Now we study another important special case of Euclidean approximation. In this particular case, we wish to approximate continuous functions from  $\mathcal{C}[a, b]$ , where  $[a, b] \subset \mathbb{R}$  denotes a compact interval.

For the approximation space, we take, for fixed  $n \in \mathbb{N}_0$ , the linear space  $\mathcal{P}_n$  of algebraic polynomials of degree at most  $n \in \mathbb{N}_0$ , where  $\dim(\mathcal{P}_n) = n+1$ , i.e., throughout this section, we regard the special case where  $\mathcal{S} = \mathcal{P}_n$  and  $\mathcal{F} = \mathcal{C}[a, b]$ .

We introduce an inner product for the function space  $\mathcal{C}[a, b]$  as follows. For a positive and integrable *weight function*  $w \in \mathcal{C}(a, b)$ , satisfying

$$\int_a^b w(x) dx < \infty,$$

the function space  $\mathcal{C}[a, b]$  is by

$$(f, g)_w = \int_a^b f(x)g(x)w(x) dx \quad \text{for } f, g \in \mathcal{C}[a, b]$$

equipped with an inner product, yielding the Euclidean norm  $\|\cdot\|_w = (\cdot, \cdot)_w^{1/2}$ , so that

$$\|f\|_w^2 = \int_a^b |f(x)|^2 w(x) dx \quad \text{for } f \in \mathcal{C}[a, b].$$

Later in this section, we make concrete examples for the weight function  $w$ .

To approximate functions from  $\mathcal{C}[a, b]$ , we apply Theorem 4.5, so that we can, for  $f \in \mathcal{C}[a, b]$ , represent the unique best approximation  $s^* \in \mathcal{P}_n$  to  $f$  explicitly. In order to do so, we need an orthogonal system for  $\mathcal{P}_n$ . To this end, we propose an algorithm, which constructs for any weighted inner product  $(\cdot, \cdot)_w$  an orthogonal basis

$$\{p_0, p_1, \dots, p_n\} \subset \mathcal{P}_n$$

for the polynomial space  $\mathcal{P}_n$ .

The following orthogonalization algorithm by *Gram-Schmidt*<sup>7</sup> belongs to the standard repertoire of linear algebra. In this iterative method, a given basis  $\mathcal{B} \subset \mathcal{S}$  of a finite-dimensional Euclidean space  $\mathcal{S}$  is, by successive orthogonal projections of the basis elements from  $\mathcal{B}$ , transformed into an orthogonal basis of  $\mathcal{S}$ . The formulation of this constructive method is detailed in the **Gram-Schmidt algorithm**, Algorithm 4, which we here apply to the monomial basis  $\mathcal{B} = \{1, x, x^2, \dots, x^n\}$  of  $\mathcal{S} = \mathcal{P}_n$ .

<sup>7</sup> ERHARD SCHMIDT (1876-1959), German mathematician

**Algorithm 4** Gram-Schmidt algorithm

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1: function GRAM-SCHMIDT
2:   let  $p_0 := 1$ ;
3:   for  $k = 0, \dots, n - 1$  do
4:     let

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$$p_{k+1} := x^{k+1} - \sum_{j=0}^k \frac{(x^{k+1}, p_j)_w}{\|p_j\|_w^2} p_j;$$

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5:   end for
6: end function

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**Proposition 4.15.** *The polynomials  $p_0, \dots, p_n \in \mathcal{P}_n$ , output by the Gram-Schmidt algorithm, Algorithm 4, form an orthogonal basis for  $\mathcal{P}_n$ .*

*Proof.* Obviously,  $p_k \in \mathcal{P}_k \subset \mathcal{P}_n$ , for all  $0 \leq k \leq n$ . Moreover, the orthogonality relation

$$p_{k+1} = x^{k+1} - \Pi(x^{k+1}) \perp \mathcal{P}_k \quad \text{for all } k = 0, \dots, n-1,$$

holds, where

$$\Pi(x^{k+1}) = \sum_{j=0}^k \frac{(x^{k+1}, p_j)_w}{\|p_j\|_w^2} p_j$$

is the orthogonal projection of the monomial  $x^{k+1}$  onto  $\mathcal{P}_k$  w.r.t.  $(\cdot, \cdot)_w$ . Therefore, the polynomials  $p_0, \dots, p_n$  form an orthogonal basis for  $\mathcal{P}_n$ . ■

Note that the orthogonalization method of Gram-Schmidt guarantees, for any weighted inner product  $(\cdot, \cdot)_w$ , the existence of an orthogonal basis for  $\mathcal{P}_n$  with respect to  $(\cdot, \cdot)_w$ . Moreover, the Gram-Schmidt construction of orthogonal polynomials in Algorithm 4 is unique up to  $n+1$  (non-vanishing) scaling factors, one for the initialization (in line 2) and one for each of the  $n$  for loop cycles (line 4). The scaling factors could be used to normalize the orthogonal system of polynomials, where the following options are commonly used.

- *Normalization of the leading coefficient*  
 $p_0 \equiv 1$  and  $p_k(x) = x^k + q_{k-1}(x)$  for some  $q_{k-1} \in \mathcal{P}_{k-1}$  for  $k = 1, \dots, n$
- *Normalization at one*  
 $p_k(1) = 1$  for all  $k = 0, \dots, n$
- *Normalization of norm (orthonormalization)*  
 Let  $p_0 := p_0 / \|p_0\|_w$  (line 2) and  $p_k := p_k / \|p_k\|_w$  (line 4),  $k = 1, \dots, n$ .

However, the Gram-Schmidt algorithm is problematic for numerical reasons. In fact, on the one hand, it is *unstable*, especially for input bases  $\mathcal{B}$  with almost linearly dependent basis elements. On the other hand, the Gram-Schmidt algorithm is very *inefficient*. In contrast, the following *three-term recursion* is much more suitable for efficient and stable constructions of orthogonal polynomials.

**Theorem 4.16.** *For any weighted inner product  $(\cdot, \cdot)_w$ , there are unique orthogonal polynomials  $p_k \in \mathcal{P}_k$ , for  $k \geq 0$ , with leading coefficient one. The orthogonal polynomials  $(p_k)_{k \in \mathbb{N}_0}$  satisfy the three-term recursion*

$$p_k(x) = (x + a_k)p_{k-1}(x) + b_k p_{k-2}(x) \quad \text{for } k \geq 1 \quad (4.26)$$

for initial values  $p_{-1} \equiv 0$ ,  $p_0 \equiv 1$  and coefficients

$$a_k = -\frac{(xp_{k-1}, p_{k-1})_w}{\|p_{k-1}\|_w^2} \text{ for } k \geq 1 \quad \text{and} \quad b_1 = 1, b_k = -\frac{\|p_{k-1}\|_w^2}{\|p_{k-2}\|_w^2} \text{ for } k \geq 2.$$

*Proof.* We prove the statement by induction on  $k$ .

*Initial step:* For  $k = 0$ , the constant  $p_0 \equiv 1$  is the unique polynomial in  $\mathcal{P}_0$  with leading coefficient one.

*Induction hypothesis:* Assume that  $p_0, \dots, p_{k-1}$ ,  $k \geq 1$ , are unique orthogonal polynomials with leading coefficient one, where  $p_j \in \mathcal{P}_j$  for  $j = 0, \dots, k-1$ .

*Induction step ( $k-1 \rightarrow k$ ):* Let  $p_k \in \mathcal{P}_k \setminus \mathcal{P}_{k-1}$  be a polynomial with leading coefficient one. Then, the difference  $p_k - xp_{k-1}$  lies in  $\mathcal{P}_{k-1}$ , so that we have (with using the orthogonal basis  $p_0, \dots, p_{k-1}$  of  $\mathcal{P}_{k-1}$ ) the representation

$$p_k(x) - xp_{k-1}(x) = \sum_{j=0}^{k-1} c_j p_j(x) \quad \text{with } c_j = \frac{(p_k - xp_{k-1}, p_j)_w}{\|p_j\|_w^2}.$$

We now formulate necessary conditions for the coefficients  $c_j$ , under which the orthogonality  $p_k \perp \mathcal{P}_{k-1}$  holds. From  $p_k \perp \mathcal{P}_{k-1}$  we get

$$c_j = -\frac{(xp_{k-1}, p_j)_w}{\|p_j\|_w^2} = -\frac{(p_{k-1}, xp_j)_w}{\|p_j\|_w^2} \quad \text{for } j = 0, \dots, k-1.$$

This in turn implies  $c_0 = \dots = c_{k-3} = 0$  and, moreover,

$$c_{k-1} = -\frac{(xp_{k-1}, p_{k-1})_w}{\|p_{k-1}\|_w^2} \text{ and } c_{k-2} = -\frac{(p_{k-1}, xp_{k-2})_w}{\|p_{k-2}\|_w^2} = -\frac{(p_{k-1}, p_{k-1})_w}{\|p_{k-2}\|_w^2}.$$

Therefore, all coefficients  $c_0, \dots, c_{k-1}$  are uniquely determined, whereby  $p_k$  is uniquely determined. Moreover, the stated three-term recursion in (4.26),

$$p_k(x) = (x + c_{k-1})p_{k-1}(x) + c_{k-2}p_{k-2}(x) = (x + a_k)p_{k-1}(x) + b_k p_{k-2}(x)$$

holds with  $a_k = c_{k-1}$ , for  $k \geq 1$ ,  $b_k = c_{k-2}$ , for  $k \geq 2$ , and where  $b_1 := 1$ . ■

**Remark 4.17.** Due to the uniqueness of the coefficients  $a_k$ , for  $k \geq 1$ , and  $b_k$ , for  $k \geq 2$ , the conditions of the three-term recursion (4.26) are also sufficient, i.e., the three-term recursion in (4.26) generates the unique orthogonal polynomials  $p_k \in \mathcal{P}_k$ , for  $k \geq 0$ , w.r.t. the weighted inner product  $(\cdot, \cdot)_w$ . □

Next, we discuss important properties of orthogonal polynomials, where their zeros are of particular interest. We can show that orthogonal polynomials have only simple zeros. To this end, we first prove a more general result for continuous functions.

**Theorem 4.18.** *Let  $g \in \mathcal{C}[a, b]$  satisfy  $(g, p)_w = 0$  for all  $p \in \mathcal{P}_n$ , i.e.,  $g \perp \mathcal{P}_n$ , for  $n \in \mathbb{N}_0$ . Then, either  $g$  vanishes identically on  $[a, b]$  or  $g$  has at least  $n + 1$  zeros with changing sign in  $(a, b)$ .*

*Proof.* Suppose  $g \in \mathcal{C}[a, b] \setminus \{0\}$  has only  $k < n + 1$  zeros

$$a < x_1 < \dots < x_k < b$$

with changing sign. Then, the product  $g \cdot p$  between  $g$  and the polynomial

$$p(x) = \prod_{j=1}^k (x - x_j) \in \mathcal{P}_k \subset \mathcal{P}_n$$

has no sign change on  $(a, b)$ . Therefore, the inner product

$$(g, p)_w = \int_a^b g(x)p(x)w(x) dx$$

cannot vanish. This is in contradiction to the assumed orthogonality  $g \perp \mathcal{P}_n$ . Therefore,  $g$  has at least  $n + 1$  zeros with changing sign in  $(a, b)$ . ■

**Corollary 4.19.** *Suppose  $p_n \in \mathcal{P}_n$  is a polynomial satisfying  $p_n \perp \mathcal{P}_{n-1}$ , for  $n \in \mathbb{N}$ . Then, either  $p_n \equiv 0$  or  $p_n$  has exactly  $n$  simple zeros in  $(a, b)$ .*

*Proof.* On the one hand, by Theorem 4.18,  $p_n$  has at least  $n$  pairwise distinct zeros in  $(a, b)$ . Now suppose  $p_n \not\equiv 0$ . Since  $p_n$  is an algebraic polynomial in  $\mathcal{P}_n \setminus \{0\}$ ,  $p_n$  has, on the other hand, at most  $n$  zeros. Altogether,  $p_n$  has exactly  $n$  zeros in  $(a, b)$ , where each zero must be simple. ■

**Corollary 4.20.** *Let  $p_n^* \in \mathcal{P}_n$  be a best approximation to  $f \in \mathcal{C}[a, b] \setminus \mathcal{P}_n$ . Then, the error function  $p_n^* - f$  has at least  $n + 1$  zeros with changing sign in  $(a, b)$ .*

*Proof.* According to Remark 4.2, we have the orthogonality  $p_n^* - f \perp \mathcal{P}_n$ . Since  $f \notin \mathcal{P}_n$ , the error function  $p_n^* - f$  has, due to Theorem 4.18, at least  $n + 1$  zeros with changing sign in  $(a, b)$ . ■

We remark that Corollary 4.20 yields a necessary condition to characterize the best approximation  $p^* \in \mathcal{P}_n$  to  $f \in \mathcal{C}[a, b]$ . This condition could *a posteriori* be used for consistency check. If we knew the  $n + 1$  simple zeros  $X = \{x_1, \dots, x_{n+1}\} \subset (a, b)$  of the error function  $p^* - f$  *a priori*, then we

would be able to compute the best approximation  $p^* \in \mathcal{P}_n$  via the interpolation conditions  $p_X^* = f_X$ . However, in the general case, the zeros of  $p^* - f$  are usually unknown.

We finally introduce three relevant families of orthogonal polynomials. For a more comprehensive discussion on orthogonal polynomials, we refer to the classical textbook [67] by Gábor Szegő<sup>8</sup> or, for numerical aspects, to the textbook [57].

### 4.4.1 Chebyshev Polynomials

We have studied the *Chebyshev polynomials*

$$T_n(x) = \cos(n \arccos(x)) \quad \text{for } n \in \mathbb{N}_0 \quad (4.27)$$

already in Section 2.5. Let us first recall some of the basic properties of the Chebyshev polynomials  $T_n \in \mathcal{P}_n$ , in particular the three-term recursion from Theorem 2.27,

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \quad \text{for } n \in \mathbb{N} \quad (4.28)$$

with initial values  $T_0 \equiv 1$  and  $T_1(x) = x$ .

Now we show that the Chebyshev polynomials  $\{T_0, \dots, T_n\} \subset \mathcal{P}_n$  are orthogonal polynomials w.r.t. the weight function  $w : (-1, 1) \rightarrow (0, \infty)$ , defined as

$$w(x) = \frac{1}{\sqrt{1-x^2}} \quad \text{for } x \in (-1, 1). \quad (4.29)$$

**Theorem 4.21.** *For  $n \in \mathbb{N}_0$  the set of Chebyshev polynomials  $\{T_0, \dots, T_n\}$  form an orthogonal basis for  $\mathcal{P}_n$  w.r.t. the weight function  $w$  in (4.29), where*

$$(T_j, T_k)_w = \begin{cases} 0 & \text{for } j \neq k \\ \pi & \text{for } j = k = 0 \\ \pi/2 & \text{for } j = k > 0 \end{cases} \quad \text{for } 0 \leq j, k \leq n. \quad (4.30)$$

*Proof.* By the substitution  $\phi = \arccos(x)$  we show the orthogonality

$$\begin{aligned} (T_j, T_k)_w &= \int_{-1}^1 \frac{T_j(x)T_k(x)}{\sqrt{1-x^2}} dx = \int_{-1}^1 \frac{\cos(j \arccos(x)) \cos(k \arccos(x))}{\sqrt{1-x^2}} dx \\ &= \int_{\pi}^0 \frac{\cos(j\phi) \cos(k\phi)}{\sqrt{1-\cos^2(\phi)}} (-\sin(\phi)) d\phi = \int_0^{\pi} \cos(j\phi) \cos(k\phi) d\phi \\ &= \|T_k\|_w^2 \delta_{jk} \end{aligned}$$

by using Theorem 4.11. Theorem 4.11 also yields the stated values for the squared norms  $\|T_k\|_w^2 = (T_k, T_k)_w$ . ■

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<sup>8</sup> GÁBOR SZEGŐ (1895–1985), Hungarian mathematician

We remark that the Chebyshev polynomials are normalized by

$$T_n(1) = 1 \quad \text{for all } n \geq 0.$$

Indeed, this follows directly by induction from the three-term recursion (4.28). Due to Corollary 2.28, the  $n$ -th Chebyshev polynomial  $T_n$  has, for  $n \geq 1$ , the leading coefficient  $2^{n-1}$ , and so the scaled polynomial

$$p_n(x) = 2^{1-n}T_n(x) \quad \text{for } n \geq 1 \quad (4.31)$$

has leading coefficient one. Thus, the orthogonal polynomials  $p_0, \dots, p_n \in \mathcal{P}_n$  satisfy the three-term recursion (4.26) in Theorem 4.16. We now show that the three-term recursion in (4.26) is consistent with the three-term recursion (4.28) for the Chebyshev polynomials.

To this end, we compute the coefficients  $a_k$  and  $b_k$  from Theorem 4.16. We first remark that the coefficients  $a_k$  are invariant under scalings of the basis elements  $p_k$ . Thus we can show that for the case of the Chebyshev polynomials all coefficients  $a_k$  in the three-term recursion (4.26) must vanish, since by the substitution  $\phi = \arccos(x)$  we have

$$(xT_k(x), T_k(x))_w = \int_0^\pi \cos(\phi) \cos^2(k\phi) d\phi = 0 \quad \text{for all } k \geq 0$$

for the nominator of  $a_{k+1}$  in (4.26). For the coefficients  $b_k$ , we get  $b_1 = 1$ ,  $b_2 = -1/2$ , and

$$b_{k+1} = -\frac{\|p_k\|_w^2}{\|p_{k-1}\|_w^2} = -\frac{\|2^{1-k}T_k\|_w^2}{\|2^{2-k}T_{k-1}\|_w^2} = -\frac{1}{4} \frac{\|T_k\|_w^2}{\|T_{k-1}\|_w^2} = -\frac{1}{4} \quad \text{for } k \geq 2.$$

From Theorem 4.16, we obtain  $p_0 \equiv 1$ ,  $p_1(x) = x$ ,  $p_2(x) = x^2 - 1/2$  and

$$p_{k+1}(x) = xp_k(x) - \frac{1}{4}p_{k-1}(x) \quad \text{for } k \geq 2.$$

Rescaling with (4.31) finally yields the three-term recursion (4.28).

In our above computations, we rely on structural advantages of Chebyshev polynomials: On the one hand, the *degree-independent* representation of the squared norms  $\|T_k\|_w^2$  in (4.30) simplifies the calculations of the coefficients  $b_k$  significantly. On the other hand, for the calculations of the coefficients  $a_k$  we can take advantage of the orthonormality of the even trigonometric polynomials  $\cos(k\cdot)$ . We wish to further discuss this important relation between the orthonormal system  $(\cos(k\cdot))_{k \in \mathbb{N}_0}$  and the orthogonal system  $(T_k)_{k \in \mathbb{N}_0}$ .

By our previous (more general) investigations in Section 4.2 the unique best approximation  $p_n^* \in \mathcal{P}_n$  to a function  $f \in \mathcal{C}[-1, 1]$  is given by the orthogonal projection  $\Pi_n f \equiv \Pi_{\mathcal{P}_n} f$  of  $f$  onto  $\mathcal{P}_n$ ,

$$\Pi_n f = \sum_{k=0}^n \frac{(f, T_k)_w}{\|T_k\|_w^2} T_k = \frac{1}{\pi}(f, 1)_w + \frac{2}{\pi} \sum_{k=1}^n (f, T_k)_w T_k, \quad (4.32)$$

where the form of **Chebyshev partial sum** in (4.32) reminds us on the form of the Fourier partial sum  $F_n f$  from Corollary 4.12. Indeed, the coefficients in the series expansion for the best approximation  $\Pi_n f$  in (4.32) can be identified as Fourier coefficients.

**Theorem 4.22.** *For  $f \in \mathcal{C}[-1, 1]$  the coefficients of the Chebyshev partial sum (4.32) coincide with the Fourier coefficients  $a_k \equiv a_k(g)$  of the even function  $g(x) = f(\cos(x))$ , so that*

$$\Pi_n f = \frac{a_0}{2} + \sum_{k=1}^n a_k T_k. \quad (4.33)$$

*Proof.* For  $f \in \mathcal{C}[-1, 1]$ , the coefficients  $(f, T_k)_w$  in (4.32) can be computed by using the substitution  $\phi = \arccos(x)$ :

$$\begin{aligned} (f, T_k)_w &= \int_{-1}^1 \frac{f(x) T_k(x)}{\sqrt{1-x^2}} dx = \int_0^\pi f(\cos(\phi)) \cos(k\phi) d\phi \\ &= \frac{\pi}{2} \frac{1}{\pi} \int_0^{2\pi} f(\cos(x)) \cos(kx) dx = \frac{\pi}{2} a_k(g), \end{aligned}$$

where  $a_k(g)$ ,  $k \geq 1$ , denotes the  $k$ -th Fourier coefficient of  $g(x) = f(\cos(x))$ . For  $k = 0$  we finally get the Fourier coefficient  $a_0$  of  $g$  by

$$\frac{(f, T_0)_w}{\|T_0\|_w^2} = \frac{\pi}{2} a_0(g) \frac{1}{\pi} = \frac{a_0(g)}{2}.$$

■

As we had remarked at the end of Section 4.3, the Fourier coefficients  $a_k \equiv a_k(g)$  can efficiently be approximated by the fast Fourier transform (FFT). Now we introduce the *Clenshaw algorithm* [15], Algorithm 5, which, on input coefficients  $a = (a_0, \dots, a_n)^T \in \mathbb{R}^{n+1}$ , yields an efficient and stable evaluation of the Chebyshev partial sum (4.33) at  $x \in [-1, 1]$ .

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#### Algorithm 5 Clenshaw algorithm

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1: function CLENSHAW( $a, x$ )
2:   Input: coefficients  $a = (a_0, \dots, a_n)^T \in \mathbb{R}^{n+1}$  and  $x \in [-1, 1]$ .
3:
4:   let  $z_{n+1} := 0$ ;  $z_n := a_n$ ;
5:   for  $k = n - 1, \dots, 0$  do
6:     let  $z_k := a_k + 2xz_{k+1} - z_{k+2}$ ;
7:   end for
8:   return  $(\Pi_n f)(x) = (z_0 - z_2)/2$ .
9: end function

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To verify the Clenshaw algorithm, we use the recursion for the Chebyshev polynomials in (4.28). By the assignment in line 6 of the Clenshaw algorithm, we get the representation

$$a_k = z_k - 2xz_{k+1} + z_{k+2} \quad \text{for } k = n-1, \dots, 0 \quad (4.34)$$

for the coefficients of the Chebyshev partial sum, where for  $k = n$  with  $z_{n+1} = 0$  and  $z_n = a_n$  we get  $z_{n+2} = 0$ . The sum over the last  $n$  terms of the Chebyshev partial sum (4.33) can be rewritten by using the representation in (4.34) in combination with the recursion (4.28):

$$\begin{aligned} \sum_{k=1}^n a_k T_k(x) &= \sum_{k=1}^n (z_k - 2xz_{k+1} + z_{k+2}) T_k(x) \\ &= \sum_{k=1}^n z_k T_k(x) - \sum_{k=2}^{n+1} 2xz_k T_{k-1}(x) + \sum_{k=3}^{n+2} z_k T_{k-2}(x) \\ &= z_1 T_1(x) + z_2 T_2(x) - 2xz_2 T_1(x) \\ &\quad + \sum_{k=3}^n z_k [T_k(x) - 2xT_{k-1}(x) + T_{k-2}(x)] \\ &= z_1 x + z_2 (2x^2 - 1) - 2xz_2 x \\ &= z_1 x - z_2. \end{aligned}$$

With  $a_0 = z_0 - 2xz_1 + z_2$  we get the representation

$$(H_n f)(x) = \frac{a_0}{2} + \sum_{k=1}^n a_k T_k(x) = \frac{1}{2} (z_0 - 2xz_1 + z_2 + 2z_1 x - 2z_2) = \frac{1}{2} (z_0 - z_2).$$

Finally, we provide a *memory efficient* implementation of the Clenshaw algorithm, Algorithm 6.

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**Algorithm 6** Clenshaw algorithm (memory efficient)

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1: function CLENSHAW( $a, x$ )
2:   Input: coefficients  $a = (a_0, \dots, a_n)^T \in \mathbb{R}^{n+1}$  and  $x \in [-1, 1]$ .
3:
4:   let  $z \equiv (z_0, z_1, z_2) := (a_n, 0, 0)$ ;
5:   for  $k = n-1, \dots, 0$  do
6:     let  $z_2 = z_1$ ;  $z_1 = z_0$ ;
7:     let  $z_0 = a_k + 2x \cdot z_1 - z_2$ ;
8:   end for
9:   return  $(H_n f)(x) = (z_0 - z_2)/2$ .
10: end function

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### 4.4.2 Legendre Polynomials

Now we discuss another example for orthogonal polynomials on  $[-1, 1]$ .

**Definition 4.23.** For  $n \in \mathbb{N}_0$ , the Rodrigues<sup>9</sup> formula

$$L_n(x) = \frac{d^n}{dx^n} ((x^2 - 1)^n) \frac{n!}{(2n)!} \quad \text{for } n \geq 0 \quad (4.35)$$

defines the  $n$ -th Legendre<sup>10</sup> polynomial. ○

We show that the Legendre polynomials are the (unique) orthogonal polynomials with leading coefficient one, belonging to the weight function  $w \equiv 1$ . Therefore, we regard the usual (unweighted)  $L^2$  inner product on  $\mathcal{C}[-1, 1]$ , defined as

$$(f, g)_w := (f, g) = \int_{-1}^1 f(x)g(x) dx \quad \text{for } f, g \in \mathcal{C}[-1, 1].$$

**Theorem 4.24.** For  $n \in \mathbb{N}_0$ , the Legendre polynomials  $L_0, \dots, L_n$  form an orthogonal basis for  $\mathcal{P}_n$  with respect to the weight function  $w \equiv 1$  on  $[-1, 1]$ .

*Proof.* Obviously,  $L_k \in \mathcal{P}_k \subset \mathcal{P}_n$  for all  $0 \leq k \leq n$ .

Now for  $0 \leq k \leq n$  we consider the integral

$$I_{nk} = \int_{-1}^1 \frac{d^n}{dx^n} ((x^2 - 1)^n) \frac{d^k}{dx^k} ((x^2 - 1)^k) dx.$$

For  $0 \leq i \leq n$ , we have the representation

$$I_{nk} = (-1)^i \int_{-1}^1 \frac{d^{n-i}}{dx^{n-i}} ((x^2 - 1)^n) \frac{d^{k+i}}{dx^{k+i}} ((x^2 - 1)^k) dx, \quad (4.36)$$

as can be shown by induction (using integration by parts).

For  $i = n$  in (4.36), we have

$$I_{nk} = (-1)^n \int_{-1}^1 (x^2 - 1)^n \frac{d^{k+n}}{dx^{k+n}} ((x^2 - 1)^k) dx = 0 \quad \text{for } n > k \quad (4.37)$$

which implies

$$(L_n, L_k) = \frac{n!k!}{(2n)!(2k)!} I_{nk} = 0 \quad \text{for } n > k. \quad (4.38)$$

■

<sup>9</sup> BENJAMIN OLINDE RODRIGUES (1795-1851), French mathematician and banker

<sup>10</sup> ADRIEN-MARIE LEGENDRE (1752-1833), French mathematician

We note two more important properties of the Legendre polynomials.

**Theorem 4.25.** *The Legendre polynomials  $L_n$  in (4.35) satisfy the following properties.*

- (a)  $L_n$  has leading coefficient one.
- (b) We have  $L_n(-x) = (-1)^n L_n(x)$  for all  $x \in [-1, 1]$ .

*Proof.* For  $n \geq 0$ , by using (4.35), we get the representation

$$\begin{aligned} \frac{(2n)!}{n!} L_n(x) &= \frac{d^n}{dx^n} ((x^2 - 1)^n) = \frac{d^n}{dx^n} \left( \sum_{j=0}^n \binom{n}{j} x^{2j} (-1)^{n-j} \right) \\ &= n! \sum_{n/2 \leq j \leq n} \binom{n}{j} \binom{2j}{n} x^{2j-n} (-1)^{n-j}. \end{aligned} \quad (4.39)$$

(a) By (4.39) the Legendre polynomial  $L_n$  has leading coefficient one.

(b) For even  $n$ , we have that  $2j - n$  is even, and so in this case all terms in (4.39) are even, which implies that  $L_n$  is even. Likewise, we can show that  $L_n$  is odd for odd  $n$  (by analogy). Altogether, we see that statement (b) holds. ■

In conclusion, the Legendre polynomials are, for the  $L^2$  inner product  $(\cdot, \cdot)$  on  $[-1, 1]$ , the unique orthogonal polynomials with leading coefficient one. Finally, we derive a three-term recursion for the Legendre polynomials from (4.26).

**Theorem 4.26.** *The Legendre polynomials satisfy the three-term recursion*

$$L_{n+1}(x) = xL_n(x) - \frac{n^2}{4n^2 - 1} L_{n-1}(x) \quad \text{for } n \geq 1 \quad (4.40)$$

with initial values  $L_0 \equiv 1$  and  $L_1(x) = x$ .

*Proof.* Obviously,  $L_0 \equiv 1$  and  $L_1(x) = x$ .

By Theorem 4.16, the sought three-term recursion has the form (4.26), where

$$a_n = -\frac{(xL_{n-1}, L_{n-1})}{\|L_{n-1}\|^2} \quad \text{for } n \geq 1 \quad \text{and} \quad b_1 = 1, \quad b_n = -\frac{\|L_{n-1}\|^2}{\|L_{n-2}\|^2} \quad \text{for } n \geq 2.$$

By statement (b) in Theorem 4.25, the Legendre polynomial  $L_n^2$  is, for any  $n \in \mathbb{N}_0$ , even, and therefore  $xL_n^2(x)$  is odd, so that  $a_n = 0$  for all  $n \geq 0$ .

**Table 4.1.** The Legendre polynomials  $L_n$  in monomial form, for  $n = 1, \dots, 10$ .

$$L_1(x) = x$$

$$L_2(x) = x^2 - \frac{1}{3}$$

$$L_3(x) = x^3 - \frac{3}{5}x$$

$$L_4(x) = x^4 - \frac{6}{7}x^2 + \frac{3}{35}$$

$$L_5(x) = x^5 - \frac{10}{9}x^3 + \frac{5}{21}x$$

$$L_6(x) = x^6 - \frac{15}{11}x^4 + \frac{5}{11}x^2 - \frac{5}{231}$$

$$L_7(x) = x^7 - \frac{21}{13}x^5 + \frac{105}{143}x^3 - \frac{35}{429}x$$

$$L_8(x) = x^8 - \frac{28}{15}x^6 + \frac{14}{13}x^4 - \frac{28}{143}x^2 + \frac{7}{1287}$$

$$L_9(x) = x^9 - \frac{36}{17}x^7 + \frac{126}{85}x^5 - \frac{84}{221}x^3 + \frac{63}{2431}x$$

$$L_{10}(x) = x^{10} - \frac{45}{19}x^8 + \frac{630}{323}x^6 - \frac{210}{323}x^4 + \frac{315}{4199}x^2 - \frac{63}{46189}$$

Now we compute the coefficients  $b_n$  for  $n \geq 2$ .

By the representation (4.37) for the integral  $I_{nk}$  we have, for  $k = n$ ,

$$\begin{aligned} I_{nn} &= (-1)^n (2n)! \int_{-1}^1 (x^2 - 1)^n dx = (2n)! \int_{-1}^1 (1 - x^2)^n dx \\ &= (2n)! \int_{-1}^1 (1 - x)^n (1 + x)^n dx \\ &= (2n)! \cdot \frac{n!}{(n+1) \cdot \dots \cdot (2n)} \int_{-1}^1 (1 + x)^{2n} dx \\ &= (n!)^2 \cdot \left[ \frac{1}{2n+1} (1 + x)^{2n+1} \right]_{x=-1}^{x=1} \\ &= (n!)^2 \cdot \frac{2^{2n+1}}{2n+1} \end{aligned}$$

after  $n$ -fold integration by parts. This gives the representation

$$\|L_n\|^2 = \frac{(n!)^2}{((2n)!)^2} \cdot I_{nn} = \frac{(n!)^4}{((2n)!)^2} \cdot \frac{2^{2n+1}}{2n+1} \quad \text{for } n \geq 0$$

and therefore

$$\begin{aligned} b_{n+1} &= -\frac{\|L_n\|^2}{\|L_{n-1}\|^2} = -\frac{n^4}{(2n)^2(2n-1)^2} \cdot \frac{2^2(2n-1)}{2n+1} \\ &= -\frac{n^2}{(2n-1)(2n+1)} = -\frac{n^2}{4n^2-1} \quad \text{for } n \geq 1, \end{aligned}$$

which proves the stated three-term recursion. ■

The Legendre polynomials  $L_n$ , for  $n = 1, \dots, 10$ , are, in their monomial form, shown in Table 4.1. To compute the entries for Table 4.1, we have used the three-term recursion (4.40) in Theorem 4.26, with initial values  $L_0 \equiv 1$  and  $L_1(x) = x$ . In summary, we see from Theorem 4.25 that

- $L_n$  has leading coefficient one;
- $L_{2k}$  is even for  $k \in \mathbb{N}_0$ ;
- $L_{2k+1}$  is odd for  $k \in \mathbb{N}$ .

Note that the above properties of the Legendre polynomials are consistent with the representations of  $L_n$  for  $n = 2, \dots, 10$ , in Table 4.1.

#### 4.4.3 Hermite Polynomials

We finally discuss one example of orthogonal polynomials on  $\mathbb{R}$ .

**Definition 4.27.** For  $n \in \mathbb{N}_0$ , we let  $H_n : \mathbb{R} \rightarrow \mathbb{R}$ , defined as

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \quad \text{for } n \geq 0, \quad (4.41)$$

denote the  $n$ -th Hermite<sup>11</sup> polynomial. ○

Next, we show that the Hermite polynomials  $H_n$  are orthogonal polynomials on  $\mathbb{R}$  with leading coefficient  $2^n$  with respect to the weight function

$$w(x) = e^{-x^2}.$$

Therefore, in this case, we work with the weighted  $L^2$  inner product

$$(f, g)_w = (f, g) = \int_{\mathbb{R}} f(x)g(x) e^{-x^2} dx \quad \text{for } f, g \in \mathcal{C}(\mathbb{R}). \quad (4.42)$$

**Theorem 4.28.** For  $n \in \mathbb{N}_0$ , the Hermite polynomials  $H_0, \dots, H_n$  form an orthogonal basis for  $\mathcal{P}_n$  with respect to the weighted inner product  $(\cdot, \cdot)_w$ .

<sup>11</sup> CHARLES HERMITE (1822-1901), French mathematician

*Proof.* We first show for  $n \in \mathbb{N}_0$  the representation

$$w^{(n)}(x) = P_n(x) \cdot e^{-x^2} \quad \text{for some } P_n \in \mathcal{P}_n \setminus \mathcal{P}_{n-1}. \quad (4.43)$$

We prove the representation in (4.43) by induction on  $n \geq 0$ .

*Initial step:* For  $n = 0$ , we have (4.43) with  $P_0 \equiv 1 \in \mathcal{P}_0$ .

*Induction hypothesis:* Suppose the representation in (4.43) holds for  $n \in \mathbb{N}_0$ .

*Induction step* ( $n \rightarrow n+1$ ): We get the stated representation by

$$\begin{aligned} w^{(n+1)}(x) &= \frac{d}{dx} w^{(n)}(x) = \frac{d}{dx} (P_n(x) \cdot e^{-x^2}) \\ &= (P'_n(x) - 2xP_n(x)) \cdot e^{-x^2} = P_{n+1}(x) \cdot e^{-x^2} \end{aligned}$$

with  $P_{n+1}(x) = P'_n(x) - 2xP_n(x)$ , where  $P_{n+1} \in \mathcal{P}_{n+1} \setminus \mathcal{P}_n$  for  $P_n \in \mathcal{P}_n \setminus \mathcal{P}_{n-1}$ .

Due to (4.43), the Hermite polynomial  $H_n$ ,  $n \geq 0$ , has the representation

$$H_n(x) = (-1)^n e^{x^2} \cdot P_n(x) \cdot e^{-x^2} = (-1)^n P_n(x) \quad \text{for } x \in \mathbb{R},$$

so that  $H_n \in \mathcal{P}_n \setminus \mathcal{P}_{n-1}$ . Moreover, by (4.43) we have

$$w^{(n)}(x) = (-1)^n e^{-x^2} \cdot H_n(x) \quad \text{for } x \in \mathbb{R}.$$

Now we consider for fixed  $x \in \mathbb{R}$  the function  $g_x : \mathbb{R} \rightarrow \mathbb{R}$ , defined as

$$g_x(t) := w(x+t) = e^{-(x+t)^2} \quad \text{for } t \in \mathbb{R}.$$

By Taylor series expansion on the analytic function  $g_x$  around zero, we get

$$w(x+t) = g_x(t) = \sum_{k=0}^{\infty} \frac{g_x^{(k)}(0)}{k!} t^k = \sum_{k=0}^{\infty} \frac{t^k}{k!} w^{(k)}(x) = \sum_{k=0}^{\infty} \frac{t^k}{k!} (-1)^k e^{-x^2} H_k(x).$$

This yields, for the function  $h(x, t) = e^{2xt-t^2}$ , the series expansion

$$h(x, t) = w(x-t) \cdot e^{x^2} = \sum_{k=0}^{\infty} \frac{t^k}{k!} H_k(x) \quad \text{for all } x, t \in \mathbb{R}. \quad (4.44)$$

Now on the one hand we have for  $s, t \in \mathbb{R}$  the representation

$$\begin{aligned} \int_{\mathbb{R}} e^{-x^2} h(x, t) h(x, s) dx &= \int_{\mathbb{R}} e^{-x^2} e^{2x(t+s)} e^{-(t^2+s^2)} dx \\ &= \int_{\mathbb{R}} e^{-(x-(t+s))^2} e^{2ts} dx \\ &= e^{2ts} \int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi} \cdot e^{2ts} \\ &= \sqrt{\pi} \cdot \sum_{k=0}^{\infty} \frac{(2ts)^k}{k!}. \end{aligned} \quad (4.45)$$

On the other hand, with using the uniform convergence of the series for  $h(x, t)$  in (4.44), we have the representation

$$\begin{aligned} \int_{\mathbb{R}} e^{-x^2} h(x, t) h(x, s) dx &= \int_{\mathbb{R}} e^{-x^2} \left( \sum_{k=0}^{\infty} \frac{t^k}{k!} H_k(x) \right) \left( \sum_{j=0}^{\infty} \frac{s^j}{j!} H_j(x) \right) dx \\ &= \sum_{k,j=0}^{\infty} \frac{t^k s^j}{k! j!} \int_{\mathbb{R}} e^{-x^2} H_k(x) H_j(x) dx. \end{aligned} \quad (4.46)$$

By comparing the coefficients in (4.45) and (4.46), we get

$$\int_{\mathbb{R}} e^{-x^2} H_k(x) H_j(x) dx = 2^k \sqrt{\pi} k! \cdot \delta_{jk} \quad \text{for all } j, k \in \mathbb{N}_0, \quad (4.47)$$

and so in particular

$$\|H_k\|_w^2 = 2^k \sqrt{\pi} k! \quad \text{for all } k \in \mathbb{N}_0.$$

This completes our proof. ■

Now we proof a three-term recursion for the Hermite polynomials.

**Theorem 4.29.** *The Hermite polynomials satisfy the three-term recursion*

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x) \quad \text{for } n \geq 0 \quad (4.48)$$

with the initial values  $H_{-1} \equiv 0$  and  $H_0(x) \equiv 1$ .

*Proof.* Obviously, we have  $H_0 \equiv 1$ . By applying partial differentiation to the series expansion for  $h(x, t)$  in (4.44) with respect to variable  $t$  we get

$$\frac{\partial}{\partial t} h(x, t) = 2(x - t)h(x, t) = \sum_{k=1}^{\infty} \frac{t^{k-1}}{(k-1)!} H_k(x)$$

and this implies

$$\sum_{k=0}^{\infty} \frac{t^k}{k!} 2xH_k(x) - \sum_{k=0}^{\infty} 2 \frac{t^{k+1}}{k!} H_k(x) = \sum_{k=0}^{\infty} \frac{t^k}{k!} H_{k+1}(x). \quad (4.49)$$

Moreover, we have

$$\sum_{k=0}^{\infty} \frac{t^{k+1}}{k!} H_k(x) = \sum_{k=0}^{\infty} \frac{t^{k+1}}{(k+1)!} (k+1)H_k(x) = \sum_{k=0}^{\infty} \frac{t^k}{k!} kH_{k-1}(x) \quad (4.50)$$

with  $H_{-1} \equiv 0$ . Inserting (4.50) into (4.49) gives the identity

$$\sum_{k=0}^{\infty} \frac{t^k}{k!} (2xH_k(x) - 2kH_{k-1}(x)) = \sum_{k=0}^{\infty} \frac{t^k}{k!} H_{k+1}(x). \quad (4.51)$$

By comparing the coefficients in (4.51), we finally get the stated three-term recursion in (4.48) with the initial values  $H_{-1} \equiv 0$  and  $H_0 \equiv 1$ . ■

**Table 4.2.** The Hermite polynomials  $H_n$  in monomial form, for  $n = 1, \dots, 8$ .

$$\begin{aligned}
H_1(x) &= 2x \\
H_2(x) &= 4x^2 - 2 \\
H_3(x) &= 8x^3 - 12x \\
H_4(x) &= 16x^4 - 48x^2 + 12 \\
H_5(x) &= 32x^5 - 160x^3 + 120x \\
H_6(x) &= 64x^6 - 480x^4 + 720x^2 - 120 \\
H_7(x) &= 128x^7 - 1344x^5 + 3360x^3 - 1680x \\
H_8(x) &= 256x^8 - 3584x^6 + 13440x^4 - 13440x^2 + 1680
\end{aligned}$$

By Theorem 4.29, we get another recursion for the Hermite polynomials.

**Corollary 4.30.** *The Hermite polynomials  $H_n$  satisfy the recursion*

$$H'_n(x) = 2nH_{n-1}(x) \quad \text{for } n \in \mathbb{N}. \quad (4.52)$$

*Proof.* Differentiation of  $H_n$  in (4.41) yields

$$H'_n(x) = \frac{d}{dx} \left( (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \right) = 2xH_n(x) - H_{n+1}(x),$$

whereby (4.52) follows from the three-term recursion for  $H_{n+1}$  in (4.48). ■

Further properties of the Hermite polynomials  $H_n$  follow immediately from the recursions in Theorem 4.29 and in Corollary 4.30.

**Corollary 4.31.** *The Hermite polynomials  $H_n$  in (4.41) satisfy the following properties.*

- (a)  $H_n$  has leading coefficient  $2^n$ , for  $n \geq 0$ .
- (b)  $H_{2n}$  is even and  $H_{2n+1}$  is odd, for  $n \geq 0$ .

*Proof.* Statement (a) follows by induction from the three-term recursion (4.48), whereas statement (b) follows from (4.52) with  $H_0 \equiv 1$  and  $H_1(x) = 2x$ . ■

We can conclude that for the weighted  $L^2$  inner product  $(\cdot, \cdot)_w$  in (4.42) the Hermite polynomials  $H_n$  are the unique orthogonal polynomials with leading coefficient  $2^n$ . The Hermite polynomials  $H_n$  are, for  $n = 1, \dots, 8$ , shown in their monomial form in Table 4.2.

## 4.5 Exercises

**Exercise 4.32.** Let  $\mathcal{F} = \mathcal{C}[-1, 1]$  be equipped with the Euclidean norm  $\|\cdot\|_2$ , defined by the inner product

$$(f, g) = \int_{-1}^1 f(x)g(x) \, dx \quad \text{for } f, g \in \mathcal{C}[-1, 1],$$

so that  $\|\cdot\|_2 = (\cdot, \cdot)^{1/2}$ . Compute on given coefficients  $a, b, c, d \in \mathbb{R}$ ,  $a \neq 0$ , of a cubic polynomial

$$f(x) = ax^3 + bx^2 + cx + d \in \mathcal{P}_3 \setminus \mathcal{P}_2 \quad \text{for } x \in [-1, 1]$$

the unique best approximation  $p_2^*$  to  $f$  from  $\mathcal{P}_2$  with respect to  $\|\cdot\|_2$ .

**Exercise 4.33.** Compute for  $n \in \mathbb{N}_0$  the Fourier coefficients  $a_0, \dots, a_n \in \mathbb{R}$  and  $b_1, \dots, b_n \in \mathbb{R}$  of the Fourier partial sum

$$F_n(x) = \frac{a_0}{2} + \sum_{j=1}^n [a_j \cos(jx) + b_j \sin(jx)] \quad \text{for } x \in [0, 2\pi)$$

(a) for the *rectangular wave*

$$R(x) = \begin{cases} 0 & \text{for } x \in \{0, \pi, 2\pi\} \\ 1 & \text{for } x \in (0, \pi) \\ -1 & \text{for } x \in (\pi, 2\pi); \end{cases}$$

(b) for the *saw-tooth function*

$$S(x) = \begin{cases} 0 & \text{for } x \in \{0, 2\pi\} \\ \frac{1}{2}(\pi - x) & \text{for } x \in (0, 2\pi). \end{cases}$$

Plot the graphs of  $R$  and the best approximation  $F_{10}R$  to  $R$  in one figure.

Plot the graphs of  $S$  and the best approximation  $F_{10}S$  to  $S$  in one figure.

**Exercise 4.34.** Approximate the function  $f(x) = 2x - 1$  on the unit interval  $[0, 1]$  by a trigonometric polynomial of the form

$$T_n(x) = \frac{c_0}{2} + \sum_{k=1}^n c_k \cos(k\pi x) \quad \text{for } x \in [0, 1]. \quad (4.53)$$

Compute (for arbitrary  $n \in \mathbb{N}_0$ ) the unique best approximation  $T_n^*$  of the form (4.53) to  $f$  with respect to the Euclidean norm  $\|\cdot\|_2$  on  $[0, 1]$ . Then determine the *smallest*  $m \in \mathbb{N}$ , satisfying

$$\int_0^1 |f(x) - T_m^*(x)|^2 \, dx \leq 10^{-4},$$

and give the best approximation  $T_m^*$  to  $f$  in explicit form.

**Exercise 4.35.** For a continuous, positive and integrable weight function  $w : (a, b) \rightarrow (0, \infty)$ , let  $\mathcal{C}[a, b]$  be equipped with the weighted Euclidean norm  $\|\cdot\|_w = (\cdot, \cdot)_w^{1/2}$ , defined by

$$(f, g)_w = \int_a^b f(x) g(x) w(x) dx \quad \text{for } f, g \in \mathcal{C}[a, b].$$

Moreover, let  $(p_k)_{k \in \mathbb{N}_0}$ , with  $p_k \in \mathcal{P}_k$ , be the unique sequence of orthogonal polynomials with respect to  $(\cdot, \cdot)_w$  with leading coefficient one. According to Theorem 4.16, the orthogonal polynomials  $p_k$  satisfy the three-term recursion

$$p_k(x) = (x + a_k) p_{k-1}(x) + b_k p_{k-2}(x) \quad \text{for } k \geq 1$$

with initial values  $p_{-1} \equiv 0$ ,  $p_0 \equiv 1$ , and with the coefficients

$$a_k = -\frac{(x p_{k-1}, p_{k-1})_w}{\|p_{k-1}\|_w^2} \quad \text{for } k \geq 1 \quad \text{and} \quad b_1 = 1, \quad b_k = -\frac{\|p_{k-1}\|_w^2}{\|p_{k-2}\|_w^2} \quad \text{for } k \geq 2.$$

Prove the following statements for  $k \in \mathbb{N}_0$ .

- (a) Among all polynomials  $p \in \mathcal{P}_k$  with leading coefficient one, the orthogonal polynomial  $p_k$  is *norm-minimal* with respect to  $\|\cdot\|_w$ , i.e.,

$$\|p_k\|_w = \min \{ \|p\|_w \mid p \in \mathcal{P}_k \text{ with } p(x) = x^k + q(x) \text{ for } q \in \mathcal{P}_{k-1} \}.$$

- (b) For all  $x, y \in [a, b]$ , where  $x \neq y$ , we have

$$\sum_{j=0}^k \frac{p_j(x) p_j(y)}{\|p_j\|_w^2} = \frac{1}{\|p_k\|_w^2} \frac{p_{k+1}(x) p_k(y) - p_k(x) p_{k+1}(y)}{x - y}$$

and, moreover,

$$\sum_{j=0}^k \frac{(p_j(x))^2}{\|p_j\|_w^2} = \frac{p'_{k+1}(x) p_k(x) - p'_k(x) p_{k+1}(x)}{\|p_k\|_w^2} \quad \text{for all } x \in [a, b].$$

- (c) Conclude from (b) that all zeros of  $p_k$  are simple. Moreover, conclude that  $p_{k+1}$  and  $p_k$  have no common zeros.

**Exercise 4.36.** Show the following identities of the Chebyshev polynomials.

- (a)  $T_k \cdot T_\ell = \frac{1}{2} (T_{k+\ell} + T_{|k-\ell|})$  for all  $k, \ell \in \mathbb{N}_0$ .  
 (b)  $T_k(-x) = (-1)^k T_k(x)$  for all  $k \in \mathbb{N}_0$ .  
 (c)  $T_k \circ T_\ell = T_{k\ell}$  for all  $k, \ell \in \mathbb{N}_0$ .

**Exercise 4.37.** In this problem, make use of the results in Exercise 4.36.

(a) Prove for  $g \in \mathcal{C}[-1, 1]$  and  $h(x) = x \cdot g(x)$ , for  $x \in [-1, 1]$ , the relation

$$c_0(h) = c_1(g) \quad \text{and} \quad c_k(h) = \frac{1}{2} (c_{k-1}(g) + c_{k+1}(g)) \quad \text{for all } k \geq 1$$

between the Chebyshev coefficients  $c_k(g)$  of  $g$  and  $c_k(h)$  of  $h$ .

(b) Conclude from the relation in Exercise 4.36 (c) the representation

$$T_{2k}(x) = T_k(2x^2 - 1) \quad \text{for all } x \in [-1, 1] \text{ and } k \in \mathbb{N}_0. \quad (4.54)$$

(c) Can the representation in (4.54) be used to simplify the evaluation of a Chebyshev partial sum for an *even* function in the Clenshaw algorithm, Algorithm 5? If so, how could this simplification be used for the implementation of the Clenshaw algorithm?

**Exercise 4.38.** On given coefficient functions  $a_k \in \mathcal{C}[a, b]$ , for  $k \geq 1$ , and  $b_k \in \mathcal{C}[a, b]$ , for  $k \geq 2$ , let  $p_k \in \mathcal{C}[a, b]$ , for  $k \geq 0$ , be a function sequence satisfying the three-term recursion

$$p_{k+1}(x) = a_{k+1}(x) p_k(x) + b_{k+1}(x) p_{k-1}(x) \quad \text{for } k \geq 1$$

with initial functions  $p_0 \in \mathcal{C}[a, b]$  and  $p_1 = a_1 p_0 \in \mathcal{C}[a, b]$ . Show that the sum

$$f_n(x) = \sum_{j=0}^n c_j p_j(x) \quad \text{for } x \in [a, b]$$

can, on given coefficients  $c = (c_0, \dots, c_n)^T \in \mathbb{R}^{n+1}$ , be evaluated by the following generalization of the *Clenshaw algorithm*, Algorithm 7.

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**Algorithm 7** Clenshaw algorithm

---

```

1: function CLENSHAW( $c, x$ )
2:   Input: coefficients  $c = (c_0, \dots, c_n)^T \in \mathbb{R}^{n+1}$  and  $x \in [a, b]$ .
3:
4:   let  $z_{n+1} = 0$ ;  $z_n = c_n$ ;
5:   for  $k = n - 1, \dots, 0$  do
6:     let  $z_k = c_k + a_{k+1}(x) z_{k+1} + b_{k+2}(x) z_{k+2}$ ;
7:   end for
8:   return  $f_n(x) = p_0(x) z_0$ .
9: end function

```

---

Which algorithm results especially for the evaluation of a *Legendre partial sum*

$$f_n(x) = \sum_{j=0}^n c_j L_j(x) \quad \text{for } x \in [-1, 1]$$

with the Legendre polynomials  $L_0, \dots, L_n$  (from Definition 4.23)?

**Exercise 4.39.** In this problem, we consider approximating the exponential function  $f(x) = e^{-x}$  on the interval  $[-1, 1]$  by polynomials from  $\mathcal{P}_n$ , for  $n \in \mathbb{N}_0$ , with respect to the weighted norm  $\|\cdot\|_w = (\cdot, \cdot)_w^{1/2}$ , where

$$w(x) = \frac{1}{\sqrt{1-x^2}} \quad \text{for } x \in (-1, 1).$$

To this end, we use the Chebyshev polynomials  $T_k(x) = \cos(k \arccos(x))$ . Compute the coefficients  $c^* = (c_0^*, \dots, c_n^*)^T \in \mathbb{R}^{n+1}$  of the best approximation

$$p_n^*(x) = \sum_{k=0}^n c_k^* T_k(x) \in \mathcal{P}_n \quad \text{for } x \in [-1, 1] \text{ and } n \in \mathbb{N}_0.$$

**Exercise 4.40.** In this problem, we use the Legendre polynomials

$$L_k(x) = \frac{d^k}{dx^k} ((x^2 - 1)^k) \frac{k!}{(2k)!} \quad \text{for } 0 \leq k \leq n$$

to determine the best approximation  $p_n^* \in \mathcal{P}_n$ ,  $n \in \mathbb{N}_0$ , to the exponential function  $f(x) = e^{-x}$  on  $[-1, 1]$  w.r.t. the (unweighted) Euclidean norm  $\|\cdot\|_2$ .

Compute the first eight coefficients  $c^* = (c_0^*, \dots, c_7^*)^T \in \mathbb{R}^8$  of the sought best approximation

$$p_n^*(x) = \sum_{k=0}^n c_k^* L_k(x) \quad \text{for } x \in [-1, 1].$$

**Exercise 4.41.** In this programming problem, we compare the two approximations to  $f(x) = e^{-x}$  from the previous Exercises 4.39 and 4.40.

- (a) Evaluate the two best approximations  $p_n^* \in \mathcal{P}_n$  (from Exercises 4.39 and 4.40, respectively) for  $n = 3, 4, 5, 6, 7$  at  $N + 1$  equidistant points

$$x_j = -1 + \frac{2j}{N} \quad \text{for } j = 0, \dots, N$$

for a suitable  $N \geq 1$  by the modified Clenshaw algorithm, Algorithm 6.

Plot the graphs of the functions  $p_n^*$  and  $f$  in one figure, for  $n = 3, 4, 5, 6, 7$ .

- (b) Record for your computations in (a) the approximation errors

$$\varepsilon_2 = \sqrt{\sum_{j=0}^N |p_n^*(x_j) - f(x_j)|^2} \quad \text{and} \quad \varepsilon_\infty = \max_{0 \leq j \leq N} |p_n^*(x_j) - f(x_j)|,$$

for  $n = 3, 4, 5, 6, 7$ . Display your results in one table.

- (c) Compare the approximation by Chebyshev polynomials (Exercise 4.39) with the approximation by Legendre polynomials (Exercise 4.40). Take notes of your numerical observations. Did the observed numerical results match your perception?

**Exercise 4.42.** Consider for  $n \in \mathbb{N}_0$  the *Hermite function*

$$h_n(x) := H_n(x) \cdot e^{-x^2/2} \quad \text{for } x \in \mathbb{R}, \quad (4.55)$$

where  $H_n$  denotes the  $n$ -th Hermite polynomial in (4.41).

Show that the Hermite functions  $h_n$  satisfy the differential equation

$$h_n''(x) - (x^2 - 2n - 1) h_n(x) = 0 \quad \text{for } n \geq 0.$$

Moreover, prove the recursion

$$h_{n+1}(x) = xh_n(x) - h_n'(x) \quad \text{for } n \geq 0.$$

*Hint:* Use the recursions from Theorem 4.29 and Corollary 4.30.